

The two notions of subideal-trace are different

Automated mathematics run `fa_banach_001`

Lane 14, model GPT5.6

9 August 2026

Status: candidate full solution, likely valid.

1 Source question

Let J be a two-sided ideal of $B(H)$ and let $\mathcal{I}$ be a J -ideal. Patnaik and Weiss define

$$\mathcal{U}_J(H) = \{1 + A \in \mathcal{U}(H) : A \in J\}$$

and call a linear functional on $\mathcal{I}$ a subideal-trace when it is invariant under conjugation by $\mathcal{U}_J(H)$. They later introduce the larger normalizing class

$$\mathcal{U}^{\mathcal{I}}(H) = \{U \in \mathcal{U}(H) : UXU^* \in \mathcal{I} \text{ for every } X \in \mathcal{I}\}.$$

On page 9, after Definition 5.11, they ask whether the resulting inclusion of trace classes can be proper [1].

Definition 5.11. For a J -ideal $\mathcal{I}$ and $\mathcal{U}^{\mathcal{I}}(H) := \{U \in \mathcal{U}(H) \mid UXU^* \in \mathcal{I} \text{ for } X \in \mathcal{I}\}$, a linear functional

$$\tau : \mathcal{I} \rightarrow \mathbb{C}$$

is called a $\mathcal{U}^{\mathcal{I}}(H)$ -subideal-trace if $\tau(X) = \tau(UXU^*)$ for every $X \in \mathcal{I}$ and $U \in \mathcal{U}^{\mathcal{I}}(H)$, that is, τ is $Ad_{\mathcal{U}^{\mathcal{I}}(H)}$ -invariant.

The following inclusion holds for a subideal $\mathcal{I}$:

$$Ad_{\mathcal{U}^{\mathcal{I}}(H)}\text{-invariant subideal-traces of } \mathcal{I} \subset Ad_{\mathcal{U}_J(H)}\text{-invariant subideal-traces of } \mathcal{I}$$

The next natural question is whether or not these inclusions are proper. In particular, do Definition 5.3 and Definition 5.11 define different classes of functionals on a subideal that is not a $B(H)$ -ideal? When $\mathcal{I}$ is a $B(H)$ -ideal, Remark 5.4 tells us that they are the same class.

In particular, they ask whether Definitions 5.3 and 5.11 give different classes of functionals on a subideal which is not a $B(H)$ -ideal.

2 Background from the source

For $S \in J$, the principal linear J -ideal has the form

$$(S)_J = \mathbb{C}S + JS + SJ + J(S)J. \tag{1}$$

When $(S)_J$ is not a $B(H)$ -ideal, the coefficient of S in this expression is unique. Consequently

$$\tau(\alpha S + X) = \alpha, \quad X \in JS + SJ + J(S)J, \tag{2}$$

is a well-defined linear functional, and Example 5.5 of the source proves that it is $\text{Ad}_{\mathcal{U}_J(H)}$ -invariant [1].

We also use the Fong–Radjavi criterion quoted on source PDF page 3. If T is a compact operator of infinite rank, the principal $K(H)$ -ideal generated by T is a $B(H)$ -ideal if and only if, for some integer $k > 1$,

$$s_{kn}(|T|) = o(s_n(|T|)). \quad (3)$$

3 Proof intuition

The functional in (2) remembers only the scalar coefficient of its chosen generator. Conjugation by a unitary of the form $1 + J$ changes the generator only by terms in the remainder of (1), so this coefficient cannot change. We choose a generator whose spectrum comes in opposite pairs. A block-swap unitary then normalizes the generated subideal but sends the generator to its negative, forcing the remembered coefficient to change sign.

4 Construction and proof

Let $H_0 = \ell^2(\mathbb{N})$, let $H = H_0 \oplus H_0$, and set

$$D = \text{diag}\left(1, \frac{1}{2}, \frac{1}{3}, \dots\right), \quad S = D \oplus (-D), \quad J = K(H), \quad \mathcal{I} = (S)_J.$$

Define the block-swap unitary

$$W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad W(x, y) = (y, x).$$

Theorem 1. *The ideal $\mathcal{I}$ is a J -ideal which is not a $B(H)$ -ideal. The coefficient functional τ from (2) is $\text{Ad}_{\mathcal{U}_J(H)}$ -invariant but is not $\text{Ad}_{\mathcal{U}^\mathcal{I}(H)}$ -invariant. Hence the inclusion displayed after Definition 5.11 of [1] is proper.*

Proof. The singular-value sequence of S is

$$s(S) = \left(1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}, \dots\right), \quad s_m(S) = \frac{1}{\lceil m/2 \rceil}.$$

For every fixed integer $k > 1$,

$$\frac{s_{kn}(S)}{s_n(S)} = \frac{\lceil n/2 \rceil}{\lceil kn/2 \rceil} \longrightarrow \frac{1}{k} \neq 0.$$

Thus (3) fails for every $k > 1$. The Fong–Radjavi criterion shows that $\mathcal{I} = (S)_J$ is not a $B(H)$ -ideal.

Put

$$\mathcal{I}^0 = JS + SJ + J(S)J.$$

The principal-subideal structure theorem gives $\mathcal{I} = \mathbb{C}S \oplus \mathcal{I}^0$, so the functional

$$\tau(\alpha S + X) = \alpha \quad (X \in \mathcal{I}^0)$$

is well-defined and linear.

For completeness, let $V = 1 + A \in \mathcal{U}_J(H)$, where $A \in J$. Then

$$VSV^* - S = AS + SA^* + ASA^* \in \mathcal{I}^0.$$

Moreover $\mathcal{I}^0$ is a J -ideal, and expansion of $(1 + A)X(1 + A^*)$ shows that $V\mathcal{I}^0V^* \subseteq \mathcal{I}^0$. Hence

$$V(\alpha S + X)V^* = \alpha S + X' \quad \text{for some } X' \in \mathcal{I}^0,$$

which proves $\tau(VXV^*) = \tau(X)$ for every $X \in \mathcal{I}$. Thus τ is a subideal-trace in the sense of Definition 5.3.

On the other hand,

$$WSW^* = -S, \quad WJW^* = J.$$

Conjugation by W therefore maps the J -ideal generated by S onto the J -ideal generated by $-S$:

$$WTW^* = (-S)_J = (S)_J = \mathcal{I}.$$

In particular $W \in \mathcal{U}^{\mathcal{I}}(H)$. Nevertheless,

$$\tau(WSW^*) = \tau(-S) = -1 \neq 1 = \tau(S).$$

Therefore τ is not invariant under $\mathcal{U}^{\mathcal{I}}(H)$, proving strictness of the inclusion. $\square$

Notice also that $W \notin \mathcal{U}_J(H)$: on the infinite-dimensional antisymmetric subspace $\{(x, -x) : x \in H_0\}$, the operator $W - 1$ equals -2 . Hence $W - 1$ is not compact. The separating unitary is genuinely available only to the larger normalizer class.

5 Verification report

The proof has four independent checks.

1. The singular-value ratio tends to $1/k$, so the subideal is not a $B(H)$ -ideal.
2. Non- $B(H)$ -ideality makes the coefficient in (1) unique.
3. Conjugation by $1 + J$ preserves the coefficient and hence τ .
4. The block swap normalizes the subideal but reverses the coefficient of S .

No numerical experiment is used. The first two checks use results explicitly reproduced in the source; the last two are direct computations.

6 Novelty check and limitations

The bounded search on 9 August 2026 covered the run indexes, arXiv and web queries for the exact question and both subideal-trace definitions, and the locally parsed arXiv corpus. It found no later explicit answer. The later arXiv:2301.09425 cites the survey for subideal terminology but does not discuss this trace question.

The theorem proves existence and therefore completely answers the source's properness question. It does not classify the subideals for which the two trace classes coincide or compute either full trace space.

Human-review recommendation. Check the Fong–Radjavi criterion and the directness of $\mathbb{C}S + \mathcal{I}^0$ first. Once those source results are accepted, the normalizer computation is elementary.

References

- [1] S. Patnaik and G. Weiss, *A survey on subideals of operators and an introduction to subideal-traces*, arXiv:1303.5697 (2013), especially Example 5.5, Definition 5.11, and page 9, <https://arxiv.org/abs/1303.5697>.